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KENYA

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STAFF REPORT FOR THE 2018 ARTICLE IV CONSULTATION AND ESTABLISHMENT OF PERFORMANCE CRITERIA FOR THE SECOND REVIEW UNDER STAND-BY ARRANGEMENT —DEBT SUSTAINABILITY ANALYSIS[1]

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Risk of external debt distress:	Moderate
Augmented by significant risks stemming from domestic public and/or private external debt?	No

Kenya's risk of external debt distress has increased from low to moderate. This reflects the breach of three external debt indicators—external debt service-to-export ratio, external debt service-to-revenue ratio, and the present value of external debt to export ratio—for an extended period of time under the most extreme shock. The first indicator is also breached under the baseline scenario, pointing to rising refinancing risks, although such a breach is temporary and marginal. It is related to a bullet Eurobond repayment, against the background of Kenya's continued access to international markets and a comfortable level of official foreign reserves. While public debt has risen in recent years, overall public–sector debt dynamics continue to be sustainable, with most new debt financing infrastructure to address bottlenecks and boost sustainable growth. However, reducing the fiscal deficit over the medium term, restricting borrowing to finance projects with high social and economic returns, and lengthening the maturity of non-concessional borrowing are essential to limit and eventually reverse the rise in public debt ratios, and reduce refinancing risks. In addition, the composition of fiscal financing between domestic and foreign sources should seek to contain risks of public external debt from export shocks while avoiding a crowding out of domestic bank credit to the private sector.

[1] Kenya's policies and institutions are classified as "strong" under the World Bank's Country Policy and Institutional Assessment (CPIA) Index (average score in 2014–16: 3.75). The relevant indicative thresholds for this category are: 50 percent for the NPV of debt-to-GDP ratio, 200 percent for the NPV of debt-to-exports ratio, 300 percent for the NPV of debt-to-revenue ratio, 25 percent for the debt service-to-exports ratio, and 22 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

BACKGROUND[2]

1. This DSA consists of two parts: external and public. The external DSA covers external debt of the central government and the central bank, as well as of the private sector; stress tests apply to public and publicly guaranteed (PPG) debt. The public DSA covers both external and domestic debt incurred or guaranteed by the central government; public domestic debt comprises central government debt. Government finance statistics are to be expanded to cover the recently created county governments. In this analysis, total public debt refers to the sum of public domestic and public external debt, but does not cover the entire public sector. The debt coverage excludes legacy debts of the pre-devolution county governments (whose size is modest), and non-guaranteed debt of state-owned enterprises.

2. Kenya's overall public debt has increased in the past two years. Gross public debt increased from 51.4 percent of GDP at end-2015 and 53.2 percent of GDP at end-2016 to about 58.0 percent at end-2017. About half of Kenya's public debt is owed to external creditors.

3. Most of Kenya's external public debt remains on concessional terms, but its commercial component has increased. Nominal public external debt at end-2017 was US$21.4 billion, equivalent to 28.5 percent of GDP.

- As of June 2017, multilateral creditors accounted for a bit more than 40 percent of external credit to Kenya as the share of debt from commercial creditors has increased. Of Kenya's bilateral debt, about two-thirds is owed to Paris Club members, while the remaining one-third is mainly semi-concessional loans from China to finance construction of the Standard Gauge Railway project (SGR).

- Kenya's commercial financing has several main elements. In 2014, Kenya issued its inaugural sovereign bonds, at 5-year and 10-year maturities, raising US$2 billion in June and a further US$750 million in December 2014.[3] Then in October 2015, Kenya contracted a two-year US$750 million syndicated loan at LIBOR plus 570 basis points, equivalent to an effective yield of 8 percent. Payment for nearly 90 percent of this syndicated loan was extended to April 2018 reflecting delays in the issuance of a planned Eurobond due to the protracted election period. More recently, in February 2018, Kenya raised an additional US$2 billion in a new sovereign bond issue; the bond has been issued in two equal tranches of 10 years at a coupon of 7.25 percent and 30 years at a coupon of 8.25 percent. At the same time, the maturity of the syndicated loan was extended to seven years.

[2] Data cited in this section come from the National Treasury's Monthly Bulletin and Quarterly Economic and Budgetary Reviews.

[3] The June 2014 issuance comprised two tranches: a five-year $500 million bond at a yield of 5.875 percent, and a 10-year $1.5 billion bond at 6.875 percent. In December 2014, Kenya added $250 million to the five-year tranche at a 5.0 percent yield and $500 million to the 10-year tranche at 5.9 percent.

Kenya: External Public Debt

	2013		2014		2015		2016	
	Billion US$	Share	Billion US$	Share	Billion US$	Share	Billion US$	Share
Multilateral creditors	6.47	60.5	7.07	49.2	7.30	46.4	7.62	40.0
Bilateral creditors	2.84	26.6	3.86	26.9	4.29	27.3	6.26	32.9
Commercial creditors	0.69	6.5	2.82	19.6	3.58	22.7	4.47	23.5
Others (supplier credits)	0.18	1.7	0.18	1.3	0.16	1.0	0.15	0.8
Total (excluding guarantees)	10.18	95.2	13.93	96.9	15.33	97.4	18.50	97.2
Publicly guaranteed debt	0.51	4.8	0.44	3.1	0.41	2.6	0.54	2.8
Total (including guarantees)	10.69	100.0	14.37	100.0	15.74	100.0	19.04	100.0

Source: Kenyan National Treasury.

4. Kenya's gross domestic public debt was KSh2,210 billion (29.5 percent of GDP) at end-2017. Domestic debt is issued mostly in the form of Treasury bonds (67 percent of domestic debt) and Treasury bills (32 percent). Commercial banks hold 54 percent of the domestic debt and nonbanks (pension and trust funds, insurance companies and others) hold 42 percent. The Central Bank of Kenya and non-residents officially hold the rest, although it is likely that non-residents hold a greater share than the data suggests as the commercial banking sector serves as a conduit for non-residents to gain exposure to Kenyan domestic debt. The average maturity profile of Kenya's outstanding government domestic debt remained at 5 years as at end December 2016.

UNDERLYING ASSUMPTIONS

5. This DSA is based on the following macroeconomic assumptions:

- Real GDP growth in the baseline was weaker in 2017 relative to 2016 outturn, reflecting mainly headwinds from a severe drought that dampened agriculture output and hydropower generation, a prolonged presidential election, which led to a wait-and-see attitude by investors as business sentiment slumped, and sluggish private sector credit growth, which, in part, has been adversely impacted by the interest rate cap. Growth is expected to recover in the medium to long term, predicated on the dissipation of two of the headwinds experienced in 2017 (adverse weather, and political uncertainty), the strengthening of global economy, and the completion of flagship infrastructure projects that are helping to address some critical supply-side bottlenecks. Further, ongoing structural reform efforts to improve the business environment, and unlock the potential in four key priority areas identified by the Government (agriculture, manufacturing, affordable housing and universal health coverage) will be supportive of growth in the medium to long term.

- The primary fiscal deficit was 5.6 percent in FY2016/17, and is projected to be brought down in the medium to longer term, consistent with the East African Community (EAC) Monetary Union

criteria of an overall fiscal deficit (including grants) of 3 percent of GDP.[4] Medium-term fiscal consolidation is expected to be underpinned by both expenditure and revenue measures. On expenditure, supported by the completion of some major flagship projects (SGR phase 1) and efforts to crowd-in private capital to finance some major infrastructure projects through PPPs will slow the pace of government spending, particularly the growth of development expenditures. Further the absence of one-off expenditures related to the elections and drought assistance will support fiscal consolidation over the medium term. On revenues, policy measures to rationalize exemptions, the new income tax bill (that seeks inter alia to broaden the tax base), administrative measures (including electronic tracking of cargo), and the projected pick-up in GDP growth should support a more robust recovery in revenues over the medium term.

- The current account deficit is estimated to have widened to 6.8 percent of GDP in 2017, mainly due to higher food and fuel imports because of the drought and rising global oil prices. Going forward, staff projects the current account deficit to start declining in 2018, supported by a rebound in exports (thanks to a broad-based recovery in the global economy, improved weather conditions, ongoing recovery in tourism following the lifting of travel warnings) and a slowdown in import growth as the pace of development expenditure (which has a high import intensity) slows.

Kenya: Selected Macroeconomic Assumptions

	2015	2016	2017	2018	2019	Long-term 1/
Real GDP Growth						
Current DSA	5.7	5.9	4.9	5.7	6.0	6.0
Previous DSA (December 2016)	5.8	5.5	6.0	6.2	6.5	6.5
Primary Fiscal Deficit (percent of GDP)						
Current DSA	4.7	4.9	4.8	2.6	0.5	0.9
Previous DSA (December 2016)	4.5	3.4	1.1	-0.5	-0.7	0.8
Non-interest Current Account (percent of GDP)						
Current DSA	5.3	3.7	5.1	4.8	4.1	2.5
Previous DSA (December 2016)	3.6	4.5	3.0	2.8	2.4	4.8

Source: IMF staff estimates.
1/ For current DSA, average 2023-37. For previous DSA, average 2022-36.

UNDERLYING ASSUMPTIONS

6. External solvency debt indicators remain below their policy-dependent debt burden thresholds under the DSA baseline scenario throughout the projection period (Figure 1). The present value (PV) of public and publicly guaranteed (PPG) external debt as a share of GDP is expected to rise to 31.4 percent in 2018 from 25.9 percent in 2017; this is higher than projected in the previous DSA. The PV of

[4] The EAC Monetary Union Protocol provides for fiscal convergence criteria, including a ceiling on the fiscal deficit (defined including grants) of 3 percent of GDP; and a ceiling on the gross public debt of 50 percent of GDP in present value terms. The other macroeconomic convergence criteria include a ceiling on headline inflation (8 percent) and a floor on reserve cover (4.5 months of imports). The fiscal plans outlined in the authorities' budget policy statement are consistent with Kenya meeting the EAC convergence criteria by 2021.

PPG external debt then gradually declines over time, remaining well below the 50 percent indicative threshold. The PV of external debt-to-exports and debt-to-revenue both remain below the indicative thresholds of 200 percent and 300 percent, respectively, over the entire projection period.

7. However, the liquidity indicators temporarily and marginally breach their thresholds under the baseline scenario. The debt service-to-exports ratio exceeds its threshold over two years by an average of 0.9 percentage points per year while the debt service-to-revenue ratio touches its threshold in one year. These reflect, in the case of both, the contracting of syndicated loans in the second half of FY2017/18 and, in the case of the former, worse-than-expected export performance in 2017.

8. Under the standard stress tests there are longer breaches in the PV of debt to exports ratio, the debt service-to-exports ratio and the debt service-to-revenue ratio relative to the threshold. Specifically, under the most extreme stress scenario, the PV of debt-to-exports breaches the threshold for 10 years by an average of more than 53 percentage points (Figure 1 and Table 2). Furthermore, the most extreme stress scenarios of the debt service-to-exports and debt service-to-revenue ratios also exceed the indicative thresholds.

9. Staff has increased the risk of external debt distress from low to moderate. This reflects the breach of three external debt indicators: external debt service-to-export ratio, external debt service-to-revenue ratio, and the PV of external debt to export ratio. While the first indicator is also breached under the baseline scenario, such a breach is only for a short period of time and is marginal, and is related to a bullet Eurobond repayment, against the background of Kenya's continued access to international markets and a comfortable level of official foreign reserves. The authorities should thus seek to reduce reliance on non-concessional borrowing to keep financing costs low. The authorities are also encouraged to refinance loans at a longer maturity to limit refinancing risks. In addition, continued commitment to fiscal consolidation will be key to maintaining debt sustainability.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

10. Under the baseline, total public debt as a share of GDP will peak in 2018 and then gradually decline over the medium term. In 2018 public sector debt is projected to rise to 63.2 percent, from 58.0 percent in 2017 and 53.2 percent in 2016 (due to a ramping up of infrastructure projects). Looking forward, the primary deficit is expected to be 2.6 percent of GDP in 2018 (due to both revenue and spending measures) and to decline to about zero in 2020–22, resulting in a decline in debt to 56.3 percent of GDP in 2022. In PV terms, the debt peaks at 60.6 percent of GDP in 2018 and falls thereafter. The PV of public debt-to-revenue ratio would ease from 299.6 percent in 2018 to 261.5 percent in 2022 (Table 3).

11. Total public debt as a share of GDP remains well below the low-income DSA public debt benchmark. Under the baseline, debt remains below the benchmark of 74 percent of GDP in PV terms, while the path of the debt-to-revenue and debt service-to-revenue ratios are projected to decline in the longer term. This benchmark is applicable for LICs whose CPIA score for quality of policies and institutions is assessed as strong and covers the entire public sector. In Kenya, public debt does not include legacy debts of the pre-devolution county governments (whose size is modest) as well as non-guaranteed debt of state-owned enterprises.

12. The alternative scenarios indicate that the PV of debt-to-GDP could breach its benchmark over the long term. Namely, under a standard scenario that keeps the primary balance unchanged from its 2017 level (corresponding to -4.8 percent of GDP) throughout the projection period (i.e., over 20 years), the PV of public debt-to-GDP would remain on a steady upwards trajectory and breaches the 74 percent benchmark in 2022. In addition, fixing the primary deficit at the 2017 level results in a significant increase in the debt-to-revenue ratio, rising above 400 percent in 2027.

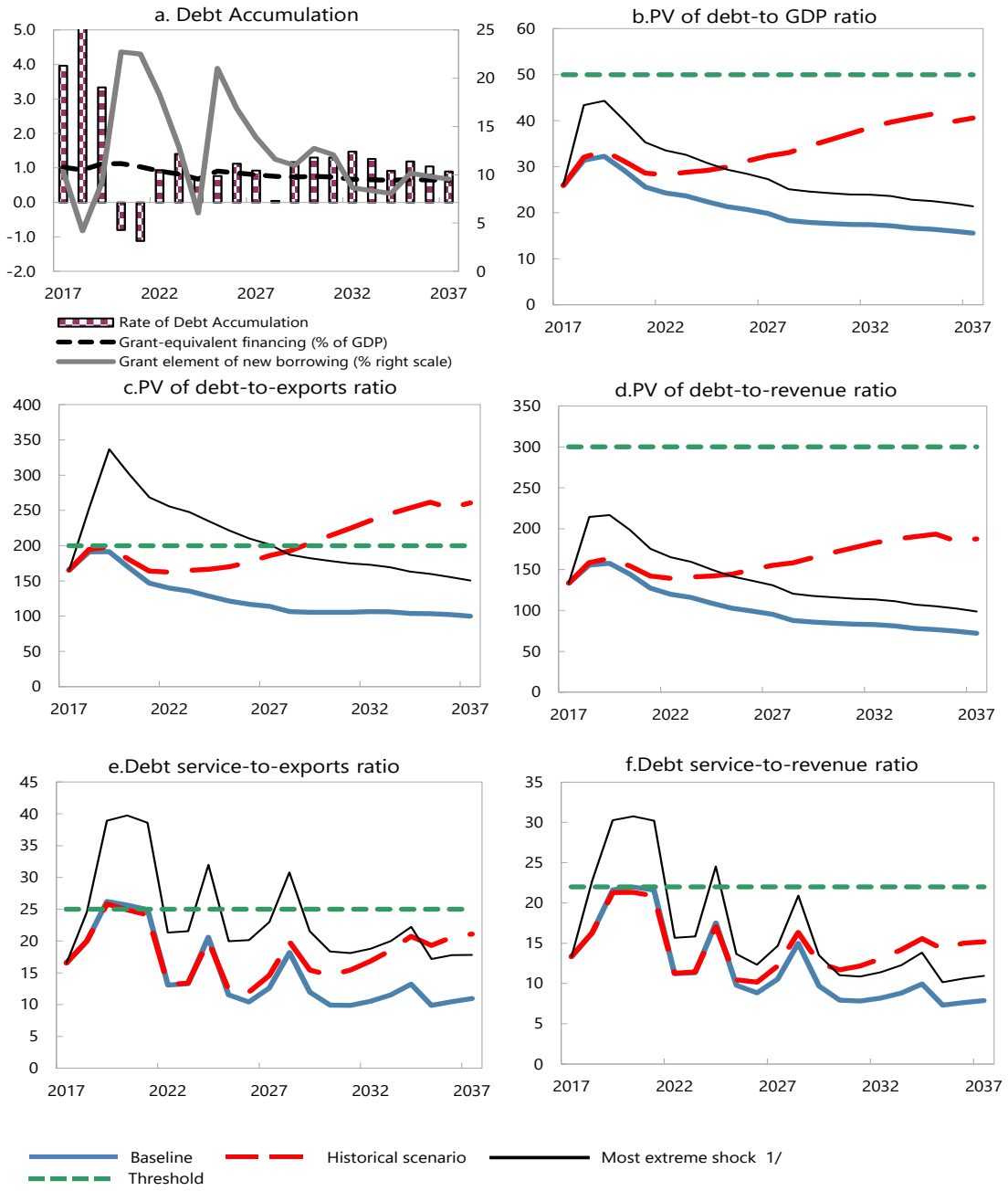
13. Further, under an extreme exchange rate shock scenario, insolvency indicators remain below thresholds. Even assuming a one-time depreciation of 30 percent taking place in 2018, the PV of debt-to-GDP would remain below its threshold. However, such a shock would bring overall public debt only about 1 percentage point of GDP away from its benchmark. Similar to the external public DSA, we note here again the vulnerability of public debt to a severe exchange rate shock.

MAIN FINDINGS AND CONCLUSIONS

14. This DSA finds that Kenya's risk of external debt distress has increased from low to moderate due to rising refinancing risks and narrower safety margins. There are breaches for three external debt indicators relative to the indicative thresholds by a significant amount and for an extended period of time under the most extreme shock. While there are breaches for one external debt indicator under the baseline scenario, such a breach is moderate in magnitude and not long lasting. In addition, the authorities are strengthening their debt management capacity to manage and prepare for large repayments on commercial borrowing. As part of this strategy, it is expected that the authorities will, over the next year, refinance loans to longer maturities. At the same time, concessional borrowing will continue to play an important role in financing investment projects due to its lower cost and longer maturity profile, while non-concessional borrowing should be limited to finance those projects with high social and economic returns. Looking ahead, such planned investment in infrastructure will be critical to raise growth and export potential, both of which will support Kenya's external debt sustainability.

15. Overall public-sector debt dynamics continue to be sustainable, although delivering on the fiscal consolidation envisaged under the program is essential to reduce risks. While the overall fiscal balance in 2016/17 was larger than projected in the previous DSA in December 2016, this was in part due to a higher execution of foreign-financed investment projects that are expected to raise growth in output and exports in the longer term. Going forward, the authorities will need to remain committed to their medium-term plans of gradual adjustment. The baseline public debt path remains consistent with the EAC convergence criteria for debt, and debt indicators are below the relevant public debt benchmark in the baseline and stress scenarios. Standard stress-testing scenarios show that if the primary deficit were to remain at current levels, public debt would remain on an upward path. This highlights the need to follow through on the intended medium-term fiscal consolidation.

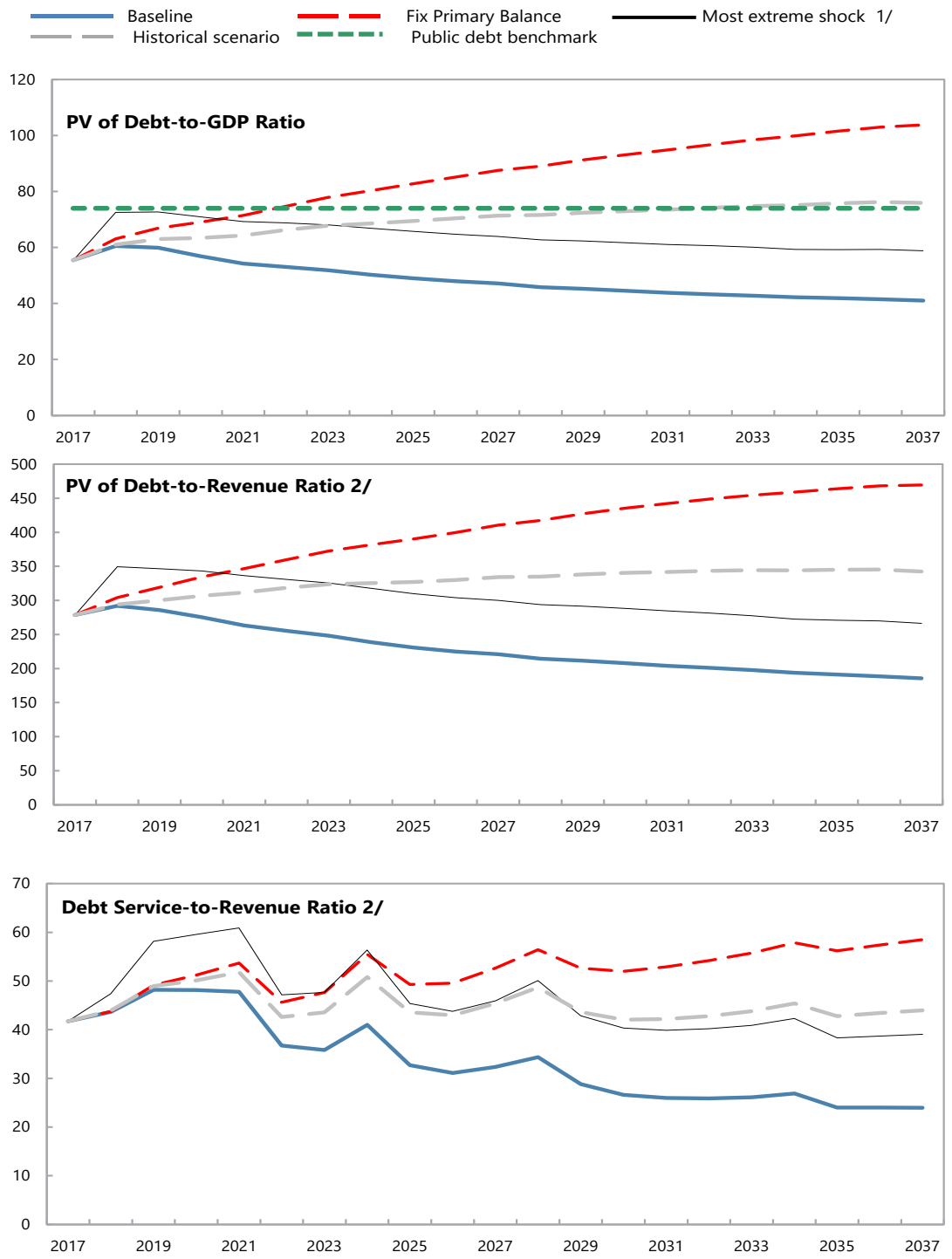
Figure 1. Kenya: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Kenya: Indicators of Public Debt Under Alternatives Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.

Table 1. Kenya: External Debt Sustainability Framework, Baseline Scenario, 2014–37 1/

(Percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**36.6**	**41.6**	**43.5**			**46.6**	**49.0**	**52.5**	**53.6**	**54.1**	**54.3**		**56.0**	**51.0**	
of which: public and publicly guaranteed (PPG)	23.5	25.7	26.4			28.5	34.1	35.2	32.3	28.8	27.5		23.0	18.3	
Change in external debt	7.3	5.0	1.9			3.1	2.4	3.6	1.1	0.5	0.2		0.2	1.1	
Identified net debt-creating flows	6.1	4.7	0.7			4.3	3.6	2.9	2.4	2.4	2.4		1.2	1.0	
Non-interest current account deficit	**9.0**	**5.3**	**3.7**	**5.9**	**2.2**	**5.1**	**4.8**	**4.1**	**3.8**	**4.0**	**4.0**		**2.6**	**2.1**	2.5
Deficit in balance of goods and services	14.7	11.1	8.7			10.1	9.5	9.0	9.0	9.1	9.1		7.7	7.4	
Exports	18.3	16.6	14.5			15.7	16.5	16.9	17.3	17.4	17.4		17.4	15.6	
Imports	33.0	27.7	23.2			25.8	25.9	25.9	26.2	26.5	26.5		25.1	22.9	
Net current transfers (negative = inflow)	-5.7	-5.4	-4.6	-5.8	0.6	-4.4	-4.3	-4.2	-4.2	-4.1	-4.1		-4.0	-4.4	-4.1
of which: official	-0.1	-0.1	0.0			0.0	-0.1	0.0	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	0.0	-0.4	-0.5			-0.5	-0.4	-0.7	-1.0	-1.1	-1.0		-1.1	-0.8	
Net FDI (negative = inflow)	**-1.2**	**-0.6**	**-0.5**	**-1.2**	**1.1**	**-0.6**	**-0.7**	**-0.7**	**-0.7**	**-0.7**	**-0.7**		**-0.6**	**-0.6**	-0.6
Endogenous debt dynamics 2/	**-1.6**	**0.0**	**-2.5**			**-0.3**	**-0.5**	**-0.5**	**-0.7**	**-0.9**	**-0.9**		**-0.7**	**-0.5**	
Contribution from nominal interest rate	1.4	1.5	1.5			1.7	1.9	2.2	2.3	2.3	2.3		2.3	2.2	
Contribution from real GDP growth	-1.4	-2.0	-2.2			-2.0	-2.4	-2.7	-3.0	-3.2	-3.2		-3.1	-2.8	
Contribution from price and exchange rate changes	-1.6	0.5	-1.8			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.1**	**0.3**	**1.2**			**-1.2**	**-1.1**	**0.6**	**-1.3**	**-2.0**	**-2.2**		**-1.0**	**0.1**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	40.8			44.0	46.3	49.6	50.4	50.9	51.1		52.9	48.3	
In percent of exports	...	...	281.7			280.3	281.2	294.0	292.1	292.0	293.4		303.4	310.2	
PV of PPG external debt	**...**	**...**	**23.7**			**25.9**	**31.4**	**32.3**	**29.1**	**25.6**	**24.3**		**19.9**	**15.6**	
In percent of exports	**...**	**...**	**163.8**			**165.4**	**191.1**	**191.4**	**168.5**	**146.9**	**139.6**		**114.0**	**100.1**	
In percent of government revenues	**...**	**...**	**129.5**			**133.5**	**155.6**	**157.8**	**144.3**	**127.2**	**119.7**		**95.2**	**72.0**	
Debt service-to-exports ratio (in percent)	**68.8**	**50.8**	**60.7**			**59.2**	**39.6**	**75.8**	**85.1**	**87.0**	**64.5**		**66.1**	**74.0**	
PPG debt service-to-exports ratio (in percent)	**16.3**	**7.6**	**9.0**			**16.5**	**19.9**	**26.2**	**25.6**	**24.9**	**13.1**		**12.6**	**10.9**	
PPG debt service-to-revenue ratio (in percent)	**15.5**	**6.8**	**7.1**			**13.3**	**16.2**	**21.6**	**22.0**	**21.6**	**11.2**		**10.5**	**7.9**	
Total gross financing need (Billions of U.S. dollars)	14.6	11.8	14.2			18.5	18.6	23.7	29.6	37.3	40.9		71.9	148.2	
Non-interest current account deficit that stabilizes debt ratio	1.7	0.3	1.8			2.0	2.4	0.5	2.7	3.5	3.8		2.3	1.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.4	5.7	5.9	5.2	2.2	4.9	5.7	6.0	6.2	6.5	6.5	6.0	6.0	6.0	6.0
GDP deflator in US dollar terms (change in percent)	5.9	-1.5	4.6	5.3	6.3	1.0	5.0	2.2	1.6	2.1	2.6	2.4	2.6	2.6	2.6
Effective interest rate (percent) 5/	5.3	4.1	4.0	3.2	1.2	4.2	4.4	5.0	4.8	4.7	4.6	4.6	4.6	4.8	4.6
Growth of exports of G&S (US dollar terms, in percent)	2.5	-5.4	-3.5	6.2	11.5	14.6	16.5	10.9	10.5	9.7	9.2	11.9	7.3	7.9	8.0
Growth of imports of G&S (US dollar terms, in percent)	10.9	-12.7	-7.0	8.2	14.4	17.4	11.8	8.0	9.4	10.1	8.9	10.9	7.7	7.9	7.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	10.3	4.2	9.2	22.7	22.5	18.3	14.5	13.9	9.5	11.5
Government revenues (excluding grants, in percent of GDP)	19.3	18.7	18.3			19.4	20.2	20.4	20.2	20.1	20.3		20.9	21.6	21.0
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.3			1.1	1.7	1.5	1.3	1.3	1.3		1.9	3.6	
of which: Grants	0.3	0.3	0.3			0.3	0.4	0.5	0.5	0.5	0.6		0.8	1.8	
of which: Concessional loans	...	...	...			0.8	1.2	1.0	0.9	0.8	0.8		1.1	1.8	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.0	0.9	1.1	1.1	1.0	0.9		0.8	0.6	0.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			17.0	9.0	15.8	34.5	35.3	32.3		27.5	26.0	25.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	61.4	64.0	70.9			75.1	83.3	90.3	97.4	105.9	115.6		175.9	408.4	
Nominal dollar GDP growth	11.5	4.2	10.7			5.9	11.0	8.3	7.9	8.7	9.2	8.5	8.8	8.8	8.8
PV of PPG external debt (in Billions of US dollars)			16.7			19.5	25.8	28.6	27.9	26.8	27.8		34.5	62.9	
(PVt-PVt-1)/GDPt-1 (in percent)						4.0	8.4	3.3	-0.8	-1.1	0.9	2.5	0.9	0.9	1.0
Gross workers' remittances (Billions of US dollars)	1.4	1.6	1.7			1.9	2.0	2.2	2.3	2.5	2.7		4.1	10.3	
PV of PPG external debt (in percent of GDP + remittances)	...	...	23.2			25.3	30.7	31.5	28.4	25.0	23.7		19.4	15.2	
PV of PPG external debt (in percent of exports + remittances)	...	...	140.0			142.7	166.5	167.4	147.9	129.3	123.1		100.5	86.1	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	7.7			14.3	17.4	22.9	22.5	21.9	11.5		11.1	9.4	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37

(Percent)

				Projections				
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to GDP ratio								
Baseline	26	31	32	29	26	24	**20**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	26	32	33	31	29	28	**32**	41
A2. New public sector loans on less favorable terms in 2017-2037 2	26	33	35	33	30	29	**27**	25
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	26	32	33	30	27	25	**21**	16
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	26	34	39	36	32	31	**24**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	26	33	35	31	28	26	**21**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	26	31	31	28	24	23	**19**	15
B5. Combination of B1-B4 using one-half standard deviation shocks	26	34	38	34	30	29	**23**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	26	43	44	40	35	34	**27**	21
PV of debt-to-exports ratio								
Baseline	165	191	191	169	147	140	**114**	100
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	165	195	198	180	164	162	**186**	261
A2. New public sector loans on less favorable terms in 2017-2037 2	165	202	209	189	171	166	**154**	164
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	165	188	188	165	145	137	**112**	98
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	165	253	337	301	269	256	**202**	150
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	165	188	188	165	145	137	**112**	98
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	165	186	182	160	140	133	**109**	97
B5. Combination of B1-B4 using one-half standard deviation shocks	165	227	269	239	211	201	**161**	130
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	165	188	188	165	145	137	**112**	98
PV of debt-to-revenue ratio								
Baseline	133	156	158	144	127	120	**95**	72
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	133	159	164	154	142	139	**155**	187
A2. New public sector loans on less favorable terms in 2017-2037 2	133	164	172	162	148	142	**129**	118
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	133	157	163	150	132	125	**99**	75
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	133	168	192	179	161	152	**117**	75
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	133	162	169	155	137	129	**102**	77
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	133	151	150	137	121	114	**91**	70
B5. Combination of B1-B4 using one-half standard deviation shocks	133	166	184	170	152	143	**111**	77
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	133	214	217	198	176	165	**131**	99

Table 2. Kenya: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37 (concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	17	20	26	26	25	13	**13**	11
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	17	20	26	25	24	13	**15**	21
A2. New public sector loans on less favorable terms in 2017-2037 2	17	20	17	14	14	13	**14**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	17	20	26	26	25	13	**13**	11
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	17	25	39	40	39	21	**23**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	17	20	26	26	25	13	**13**	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	17	20	26	25	25	13	**12**	11
B5. Combination of B1-B4 using one-half standard deviation shocks	17	23	34	34	33	18	**18**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	17	20	26	26	25	13	**13**	11
Debt service-to-revenue ratio								
Baseline	13	16	22	22	22	11	**11**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	13	16	21	21	21	11	**12**	15
A2. New public sector loans on less favorable terms in 2017-2037 2	13	16	14	12	12	11	**12**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	13	17	23	23	23	12	**11**	8
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	13	16	22	24	23	13	**13**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	13	17	24	24	24	12	**11**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	13	16	22	22	21	11	**10**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	13	17	23	24	24	13	**13**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	13	23	30	31	30	16	**15**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	6	6	6	6	6	6	**6**	6

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Kenya: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014–37

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate		Projections								
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	48.6	51.4	53.2			58.0	63.2	62.9	60.0	57.5	56.3		50.2	43.7	
of which: foreign-currency denominated	23.5	25.7	26.4			28.5	34.1	35.2	32.3	28.8	27.5		23.0	18.3	
Change in public sector debt	4.6	2.9	1.8			4.8	5.3	-0.4	-2.8	-2.6	-1.2		-0.8	-0.5	
Identified debt-creating flows	2.9	3.7	1.8			5.1	1.2	-0.8	-1.8	-2.2	-2.0		-1.2	-0.1	
Primary deficit	4.6	4.7	4.9	2.7	1.6	4.8	2.6	0.5	-0.4	-0.4	-0.1	1.2	0.7	1.8	0.9
Revenue and grants	19.8	19.1	18.7			19.9	20.7	21.0	20.7	20.6	20.8		21.3	22.1	
of which: grants	0.5	0.5	0.4			0.4	0.5	0.5	0.5	0.5	0.5		0.5	0.4	
Primary (noninterest) expenditure	24.3	23.9	23.6			25.1	24.1	22.6	21.8	21.6	21.6		22.2	23.9	
Unidentified Measures						0.3	0.8	1.1	1.5	1.3	0.9		0.2	0.0	
Automatic debt dynamics	-1.7	-1.1	-3.1			0.0	-1.3	-1.2	-1.4	-1.8	-1.9		-1.9	-1.9	
Contribution from interest rate/growth differential	-1.5	-2.0	-1.6			0.0	-1.0	-1.3	-1.5	-1.6	-1.7		-1.6	-1.7	
of which: contribution from average real interest rate	0.8	0.7	1.3			2.5	2.1	2.3	2.2	2.0	1.8		1.3	0.8	
of which: contribution from real GDP growth	-2.2	-2.6	-2.9			-2.5	-3.1	-3.6	-3.7	-3.6	-3.5		-2.9	-2.5	
Contribution from real exchange rate depreciation	-0.2	0.9	-1.5			0.0	-0.3	0.1	0.1	-0.2	-0.2		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.3	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		...	...	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.3	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.7	-0.8	0.0			-0.4	4.0	0.4	-1.0	-0.4	0.8		0.4	-0.4	
Other Sustainability Indicators															
PV of public sector debt	...	...	50.6			55.4	60.6	59.9	56.9	54.3	53.1		47.1	41.0	
of which: foreign-currency denominated	...	...	23.7			25.9	31.4	32.3	29.1	25.6	24.3		19.9	15.6	
of which: external	...	...	23.7			25.9	31.4	32.3	29.1	25.6	24.3		19.9	15.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	18.9	17.4	18.3			20.6	19.6	18.4	17.0	16.9	15.3		14.9	13.9	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	270.1			278.6	292.0	285.7	275.3	263.4	255.5		221.0	185.7	
PV of public sector debt-to-revenue ratio (in percent)	...	...	275.9			285.0	299.6	292.9	282.1	269.7	261.5		226.0	189.6	
of which: external 3/	...	...	129.5			133.5	155.6	157.8	144.3	127.2	119.7		95.2	72.0	
Debt service-to-revenue and grants ratio (in percent) 4/	39.6	32.5	35.5			41.7	43.7	48.2	48.1	47.8	36.7		32.3	24.0	
Debt service-to-revenue ratio (in percent) 4/	40.6	33.3	36.3			42.7	44.8	49.4	49.3	48.9	37.6		33.1	24.5	
Primary deficit that stabilizes the debt-to-GDP ratio	0.0	1.9	3.1			0.1	-2.7	0.8	2.5	2.2	1.1		1.5	2.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.4	5.7	5.9	5.2	2.2	4.9	5.7	6.0	6.2	6.5	6.5	6.0	6.0	6.0	6.0
Average nominal interest rate on forex debt (in percent)	2.4	2.6	3.1	1.8	0.7	3.7	4.6	4.6	4.2	4.0	4.0	4.2	3.9	4.3	4.1
Average real interest rate on domestic debt (in percent)	2.8	1.4	3.5	2.0	3.8	8.2	5.4	5.3	5.7	5.4	4.5	5.7	2.9	1.4	2.5
Real exchange rate depreciation (in percent, + indicates depreciation)	-1.2	3.9	-6.2	-2.6	7.5	0.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.1	10.0	8.1	8.9	3.6	2.7	5.3	5.8	5.3	5.1	5.1	4.9	5.1	5.1	5.1
Growth of real primary spending (deflated by GDP deflator, in percent)	12.8	3.6	4.8	2.2	4.1	11.3	1.6	-0.6	2.4	5.5	6.8	4.5	6.6	6.5	6.7
Grant element of new external borrowing (in percent)	...	...	...	...	...	10.3	4.2	9.2	22.7	22.5	18.3	14.5	13.9	9.5	...

Sources: Country authorities; and staff estimates and projections.

1/ Refers to gross debt of the central government, including CBK obligations to the IMF, excluding government deposits.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Kenya: Sensitivity Analysis for Key Indicators of Public Debt, 2017–37

(Percent)

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	55	61	60	57	54	53	47	41
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	55	61	63	63	64	66	71	76
A2. Primary balance is unchanged from 2017	55	63	67	69	71	75	88	104
A3. Permanently lower GDP growth 1/	55	62	63	61	61	61	60	68
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	55	63	67	66	66	66	65	66
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	55	62	65	64	62	61	56	49
B3. Combination of B1-B2 using one half standard deviation shocks	55	62	66	65	64	64	61	59
B4. One-time 30 percent real depreciation in 2018	55	73	73	71	69	69	64	59
B5. 10 percent of GDP increase in other debt-creating flows in 2018	55	71	71	69	68	67	60	52
PV of Debt-to-Revenue Ratio 2/								
Baseline	279	292	286	275	263	255	221	186
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	279	294	300	307	312	318	334	342
A2. Primary balance is unchanged from 2017	279	304	319	335	347	360	410	470
A3. Permanently lower GDP growth 1/	279	297	298	298	294	293	283	308
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	279	306	318	320	318	318	307	298
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	279	300	311	308	301	296	262	220
B3. Combination of B1-B2 using one half standard deviation shocks	279	301	313	313	309	307	287	266
B4. One-time 30 percent real depreciation in 2018	279	350	347	343	336	331	300	266
B5. 10 percent of GDP increase in other debt-creating flows in 2018	279	342	340	336	328	322	283	235
Debt Service-to-Revenue Ratio 2/								
Baseline	42	44	48	48	48	37	32	24
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	42	44	49	50	52	43	45	44
A2. Primary balance is unchanged from 2017	42	44	49	51	54	46	53	58
A3. Permanently lower GDP growth 1/	42	44	49	50	51	40	39	38
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	42	45	51	52	53	43	42	38
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	42	44	49	50	51	41	37	29
B3. Combination of B1-B2 using one half standard deviation shocks	42	44	50	51	52	42	40	34
B4. One-time 30 percent real depreciation in 2018	42	47	58	60	61	47	46	39
B5. 10 percent of GDP increase in other debt-creating flows in 2018	42	44	52	55	55	45	40	31

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.